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# Sarosij Bose

## Research Interests

- **Robotics/3D**: 3D Reconstruction, Scene Understanding
- **Computer Vision**: Multimodal Perception, Generative Modeling

## Education

- 2023–Present **University of California, Riverside, CA, USA**  
Ph.D. (with M.S.) in Electrical and Computer Engineering  
**Advisors**: [Amit K. Roy Chowdhury](#) & [Konstantinos Karydis](#)
- 2019–2023 **University of Calcutta, Kolkata, WB, India**  
Bachelor of Technology (BTech.) in Computer Science and Engineering  
**Advisor**: [Amlan Chakrabarti](#)  
**Thesis**: Synthetic Data Generation Using GANs and Study of Localization Techniques

## Research Experience

- Jun 2025–  
Present **Research Intern, NEC Laboratories America**, Princeton, NJ  
**Managers**: [Biplob Debnath](#) & [Ravi Rajendran](#)  
○ Working on grounding Vision-Language Models for efficient medical diagnosis.
- Sep 2023–  
Present **Graduate Research Assistant, University of California, Riverside.**  
**Advisors**: [Amit K. Roy Chowdhury](#) & [Konstantinos Karydis](#)  
○ Working on next-best view selection using uncertainty-driven sampling.  
○ Developed a novel uncertainty-driven 3D scene enhancement pipeline using camera-controllable video diffusion priors. (**ICCV 2025**)  
○ Developed self-supervised algorithms for pose estimation. (**CVPRW25, ICCV25**)
- Mar  
2023–Jun  
2023 **Research Engineer Intern, Siemens**, Bangalore, KA, India  
**Managers**: Aswin Ajayan, Deepika Kanade & Saadhana Venkataraman  
○ Worked on designing sensor realistic vehicle simulation systems.  
○ Used material properties such as **Bidirectional Reflection Distribution Function (BRDF)** for realistic object modeling.
- Jun  
2022–Sep  
2022 **MITACS Globalink Research Intern, Saint Mary's University**, Halifax, NS, Canada  
**Advisor**: [Jiju Poovvancheri](#), GSC Lab, Dept. of Maths and Computing  
○ Developed a novel algorithm for **skeleton approximation** of human models.  
○ Proposed a new sphere mesh based geometric representation.
- Jun  
2021–Aug  
2021 **Research Intern, Indian Institute of Science**, Remote  
**Advisor**: [Kunal N. Chaudhury](#), LISA Lab, EE Dept.  
○ Worked on lipschitz regularization of Convolutional Neural Networks.

## Publications/Pre-Prints

- **Uncertainty Aware Diffusion Guided Refinement of 3D Scenes** [[Paper](#)]  
Sarosij Bose, Arindam Dutta, Sayak Nag, Junge Zhang, Jiachen Li, Konstantinos Karydis, Amit K. Roy Chowdhury (*ICCV 2025*)
- **Voccl3D: A Video Benchmark Dataset for Human Pose Estimation under real Occlusions**  
Yash Garg, Saketh Bachu, Arindam Dutta, Rohit Lal, [Sarosij Bose](#), Calvin-Khang Ta, M. Salman Asif, Amit K. Roy Chowdhury (*ICCV 2025*)
- **Conformal Prediction and MLLM aided Uncertainty Quantification in Scene Graph Generation** [[Paper](#)]  
Sayak Nag, Udit Ghosh, Calvin-Khang Ta, [Sarosij Bose](#), Jiachen Li, Amit K. Roy Chowdhury (*CVPR 2025*)
- **Leveraging Synthetic Adult Datasets for Unsupervised Infant Pose Estimation** [[Paper](#)]  
[Sarosij Bose](#), Arindam Dutta, Hannah Dela Cruz, Rohit Kundu, Konstantinos Karydis, Amit K. Roy Chowdhury (*CVPRW 2025*)
- **Unsupervised Domain Adaptation for Occlusion Resilient Human Pose Estimation** [[Paper](#)]  
Arindam Dutta\*, [Sarosij Bose\\*](#), Saketh Bachu, Calvin-Khang Ta, Konstantinos Karydis, Amit K. Roy Chowdhury (*Under Review*)
- **SoccerKDNet: A Knowledge Distillation Framework for Action Recognition in Soccer Videos** [[Paper](#)] [[Code](#)]  
[Sarosij Bose](#), Saikat Sarkar, Amlan Chakrabarti  
*10th International Conference on Pattern Recognition and Machine Intelligence (PReMI), 2023*
- **Lipschitz Bound Analysis of Neural Networks** [[Paper](#)] [[Slides](#)] [[Code](#)]  
[Sarosij Bose](#)  
*13th IEEE International Conference on Computing Communication and Networking Technologies (ICCCNT), 2022*
- **A Fusion Architecture for Human Activity Recognition** [[Paper](#)] [[Slides](#)] [[Code](#)]  
[Sarosij Bose](#), Amlan Chakrabarti  
*18th IEEE India Council International Conference (INDICON), 2021*
- **ResCNN: An alternative implementation of Convolutional Neural Networks** [[Paper](#)] [[Code](#)]  
[Sarosij Bose](#), Avirup Dey  
*8th IEEE Uttar Pradesh International Conference (UPCON), 2021*

## Technical Skills

**Programming Languages:** Python, Java, C++, Javascript, MATLAB, Cuda (Basic).

**Frameworks/Tools:** PyTorch, Tensorflow, Keras, Git, Linux, HTML, FiftyOne,  $\text{\LaTeX}$ .

**Software/Libraries:** OpenCV, ROS, CVXOPT, CARLA, COLMAP, VisualSFM.

## Selected Honors and Awards

**CVPR** Recipient of full-funded travel grant award for CVPR 2025

**UCR** Recipient of the Dean's Distinguished Fellowship Award

**MITACS** Recipient of **MITACS GRI Fellowship** in Canada worth **CA\$15000**

**IIT KGP** Winner of AI EntrepreNeural 2021, sponsored by Intel and GES, IIT Kharagpur

## Volunteer Services

Served as reviewer for **ICLR, ICRA, IROS, IJCV, WACV, ICASSP, MLRC**